THE UNITED REPUBLIC OF TANZANIA.
MINISTRY OF EDUCATION SCIENCE AND TECHNOLOGY
MILITARY SCHOOL ASSOCIATION
FORM TWO PRE-NATIONAL MARKING SCHEME
PHYSICS

YEAR: AUGUST 2026

SECTION A (15 MARKS).

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	A	D	B	B	B	B	C	C	D

Marks @ 01.

2.

LIST A	i	ii	iii	iv	v
LIST B	C	A	F	H	E

Marks @ 01.

SECTION B (40 MARKS).

3 (a) Two properties of gravitational force
(i) It is always an attractive force ----- 01 mark
(ii) It is the weakest of the four fundamental force ----- 01 mark.

(b) Data given

Weight on earth (W_e) = 10,000 N
Weight on planet x (W_{Px}) = 300 N
Acceleration due to gravity on earth (g_e) = 10 N/kg
Required acceleration due to gravity on planet.

Solution.

3\. (b) From $W_e = mg_e$

$$m = \frac{W_e}{g_e}$$ ------- 01 mark

$$m = \frac{10,000N}{10N/Kg}$$ ------ 01 mark

$$m = 1000Kg$$ ------ 01 mark

Then from

$$W_{Px} = mg_{Px}$$ ------ 01 mark

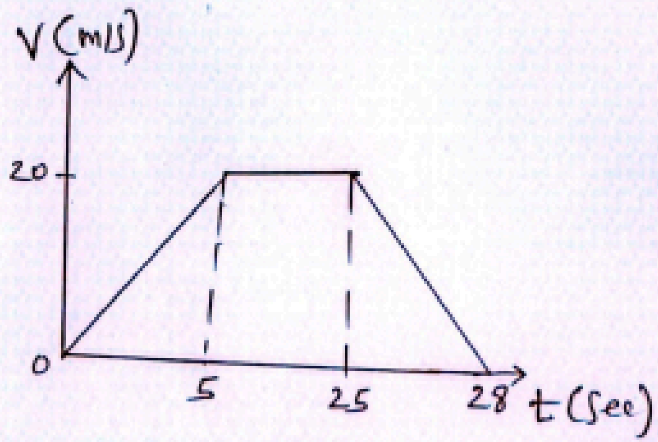
$$g_{Px} = \frac{W_{Px}}{m}$$ ------ 01 mark

$$= \frac{300N}{1000Kg}$$

$$= 0.3N/Kg$$ ---- 01 mark

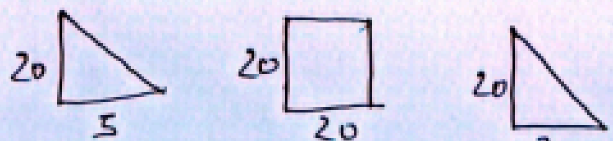
Therefore acceleration due to gravity on planet 0.3N/Kg.

4\. (a) (i)



(04 marks)

(ii)



$= \frac{1}{2} \times 20 \times 5$ $= 50m$ $\quad$ $= 20 \times 20$ $= 400m$ $\quad$ $= \frac{1}{2} \times 3 \times 20$ $= 30m$

4 (a)(ii) Total distance (d) $= 50m + 40m + 30m$
$= 480m$ (02 mark)

Therefore total distance travelled is 480m.

(b) Maximum pressure, $P_{max} = \frac{F}{A_{small}}$ --- (01 mark)

Small area (A_{small}) $= 2m \times 4m$
$= 8m^2$ -- (02 mark)

$P_{max} = \frac{100N}{8m^2}$ (03 mark)
$= 12.5N/m^2$

Therefore maximum pressure required is $12.5N/m^2$

5 (a)(i) Opaque objects are objects which do not allow light to pass through them (01 mark)

(ii) Transluscent objects are the objects which allow small amount of light to pass through (01 mark)

(iii) Transparent objects are the objects which allow all light to pass through them (01 mark)

(iv) Luminous objects are all objects which give out it's own light (01 mark).

5 (b) (i) 1st law State that "Angle of incidence is equal to the angle of reflection" (02 Marks)

2nd law state that "Incident ray, normal line and reflected ray all lie in the same plane" (02 marks)

(ii) $$n = \frac{360}{\theta} - 1 \quad \text{------ (01 mark)}$$

$$= \frac{360}{45} - 1.$$

$$= 8 - 1$$

$$= 7 \quad \text{------- (01 mark)}$$

Therefore number of Image is 7

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6 (a) (i) Hot Soup: Surface tension decrease with temperature allowing soup to spread better over the tongue enhancing taste -- (01 mark)

(ii) Ink sticks to paper, due to adhesion --(01 mark)

(iii) Kerosene in wick, due to capillarity --(01 mark)

(iv) Pond skater; due to surface tension ---(01 mark)

(b) Data given:

Mass (M) = 50kg

distance(d) = 30cm x 12 steps --- (01 mark)

= 0.3m x 12

= 3.6m.

Required Work done (W.d)

Solution

6 (b) From $W.d = F \times d$ - - - - - - - (01 mark)

$= mg \times d$ - - - - - (01 mark)

$= 50kg \times 10N/kg \times 3.6m$

$= 1800Nm$ - - - - - - (01 mark)

Therefore Work done required is 1800Nm (01 mark)

7. (a) Application of density in everyday life.

(1) It aids in the identification of gemstones such as gold and diamond. 02

(2) Density is considered during the selection of building materials 02

(3) Density is also considered during the design of swimming and diving equipment. 02

(b) Data given

Weight of object in air $W_0 = 8.0N$

weight of object in water $W_1 = 6.0N$

Weight of object in kerosene $W_2 = 6.4N$

$$\text{Relative density} = \frac{\text{upthrust in Kerosene}}{\text{upthrust in water}} \quad 01$$

$$= \frac{8.0N - 6.4N}{8.0N - 6.0N} \quad 01$$

$$= \frac{1.6}{2} \quad 01$$

$$= 0.8$$

Relative density of kerosene $= 0.8$ 01

8 (a) Three factors on which buoyant force / upthrust depends.

(1) the volume of the immersed object 01

(2) the density of the fluid in which the object is immersed; and 01

(3) the acceleration due to gravity. 01

(b) <u>Data given</u>

Volume of cork $V_{cork} = 100\,cm^3$

Density of cork $\rho_{cork} = 0.25\,g/cm^3$

Density of water $\rho_{water} = \underline{1}\,g/cm^3$

(1) $\text{Density of cork} = \dfrac{\text{Mass of cork}}{\text{Volume of cork}}$ 01

Mass of cork = Density of cork × Volume of cork 01

$= 0.25 \times 100$

$= 25\,g$ 01

According to law of floatation

weight of cork = weight of water displaced

$(Mg)_{cork} = (Mg)_{\text{water displaced}}$

$(\rho V g)_{cork} = (\rho V g)_{\text{water displaced}}$ 01

$0.25 \times 100 \times 10 = \underline{1} \times V_{\text{water displaced}} \times 10$

$\dfrac{0.25 \times 100 \times 10}{1 \times 10} = V_{\text{water displaced}}$ 01

8 (b) V water displaced = 25 cm^3

Volume of cork immersed = Volume of water displaced

= 25 cm^3 01

9. (a) A capacitor becomes charged when a voltage source is connected across its plates, causing a separating of charge electrons are removed from one plate. (Making it positive) and accumulate on the other plate (Making it negative), creating an electric field between them. 05

(b) A steel bar can be magnetised by electricity by placing it inside a solenoid and passing an electric current through the solenoid. The magnetic field produced by the current aligns the magnetic domains within the steel turning it into magnet. 05

10 (a) Parallel connection is commonly used in household wiring because it allows appliances to operate independently of each other, and each appliance receives the full main voltage. 04

(b) Data given

Emf $= 2V$

Resistor $R_1 = 6\Omega$

Resistor $R_2 = 20\Omega$

Resistor $R_3 = 10\Omega$

6Ω and 20Ω are parallel connected

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} \quad 01$$

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{6}$$

$$\frac{1}{R_T} = \frac{6+20}{120}$$

$$\frac{1}{R_T} = \frac{26}{120}$$

$$R_T = \frac{1 \times 120}{26}$$

R_T for 6Ω and $20\Omega = 4.615\Omega$

4.615Ω and 10Ω are in series connection

$$R_T = R_1 + R_2$$

$$R_T = 4.615\Omega + 10\Omega \quad 01$$

$$R_T = 14.615\Omega$$

10. (b) Voltage / Emf = 2V

Total resistance R = 14.615

The current in circuit = ?

$$V = IR \quad 01$$

$$2 = I \times 14.615$$

$$I = 0.137A \quad 01$$

Voltage at 10Ω = ?

$$V = IR$$

$$= 0.137 \times 10$$

$$= 1.37V$$

Voltage at 6Ω and 20Ω

$$= 2 - 1.37V$$

$$= 0.63V \quad 01$$

because 6Ω and 20Ω are in series Voltage at 6Ω ~~and~~ 0.63 V

$$V = IR \quad 01$$

$$0.63 = I \times 6$$

$$I = \frac{0.63}{6}$$

$$I = 0.105A \quad 01$$

10 (c) <u>Data given</u>

Total power = 100W + 15W + 300W + 1500W

$= 1975W$ 01

$P = IV$ 01

$1975 = I\,240$

$I = \frac{1975}{240}$

$I = 8.229A$ 01

The fuse will not blow off ~~because~~ it allow less current of 8.229A to pass through it. 01